ECE 329 **Homework 2 — Solutions**

1. a) (3 pts) Expressing the equation $\oint_S \mathbf{D}\cdot d\mathbf{S} = \int_V \rho dV$ in terms of its units, we write

$$\mathbf{D} = \frac{[\frac{C}{\mathrm{m}^3}]\cdot[\mathrm{m}^3]}{[m^2]} = [\frac{c}{\mathrm{m}^2}].$$

This makes sense because electric displacement gives the electric flux density. Electric flux density is a measure of the flow of electric field through a given surface. Also, we can find the same unit for $\mathbf{D}$ by using the units of $\mathbf{D} = \epsilon_o\mathbf{E} = [\frac{F}{m}]\cdot[\frac{N}{C}] = [\frac{\frac{C^2}{N\cdot m}}{m}]\cdot[\frac{N}{C}] = [\frac{C}{m^2}]$.

b) (2 pts) According to Gauss's law, the displacement flux Φ will be

$$\oint_S \mathbf{D}\cdot d\mathbf{S} = \int_V \rho dV = \int_V 3dV = 3\,\mathrm{C},$$

where its unit is "Coulomb".

c) (3 pts) For $\rho(x,y,z) = x^2+y^2+z^2\ \frac{\mathrm{C}}{\mathrm{m}^3}$,

$$\oint_S \mathbf{D}\cdot d\mathbf{S} = \int_V \rho dV = \int_{-1/2}^{1/2}\int_{-1/2}^{1/2}\int_{-1/2}^{1/2}(x^2+y^2+z^2)dxdydz = \frac{1}{4}\,\mathrm{C}.$$

d) (2 pts) Taking advantage of the symmetry of the charge distribution, it can be easily verified that the electric flux through each of the six faces of the cube must be equal, i.e., $\Phi_1 = \Phi_2 = ... = \Phi_6$. Hence, the electric flux in part (b) for any of the square surfaces of volume, V is

$$\Phi_i = \frac{1}{6}\times\frac{1}{4} = \frac{1}{24}\,\mathrm{Vm}.$$

2. Gauss' Law ($\oint_S \mathbf{D}\cdot d\mathbf{S} = Q$) is valid for any surface S enclosing the charges Q_1 and Q_2. The total flux through the $yz-$plane will be the sum of the flux through this plane due to Q_1 and Q_2. Referring to the logic based on symmetry of the charges in the Example in Lecture 3, the flux through the $yz-$plane will be the superposition of half of the flux emanating from each charge. Hence, the total flux through the entire $yz-$plane will be (4 pts)

$$\int_{yz-\mathrm{plane}} \mathbf{D}_1|_{x=0}\cdot(-\hat{x}dydz) + \mathbf{D}_2|_{x=0}\cdot(\hat{x}dydz) = -\frac{Q_1}{2}+\frac{Q_2}{2},$$

from which we obtain

$$\frac{Q_2-Q_1}{2} = 4\,\mathrm{C}. \tag{1}$$

Following the same procedure, the total flux crossing the plane located at $y=1$ in the $+\hat{y}$ direction will be (4 pts)

$$\int_{xz-\mathrm{plane}} \mathbf{D}_1|_{y=1}\cdot(\hat{y}dxdz) + \int_{xz-\mathrm{plane}} \mathbf{D}_2|_{y=1}\cdot(\hat{y}dxdz) = \frac{Q_1}{2}+\frac{Q_2}{2}$$

from which we obtain

$$\frac{Q_2+Q_1}{2} = -2\,\mathrm{C}. \tag{2}$$

Solving for (1) and (2), we obtain $Q_1 = -6\,\mathrm{C}$, and $Q_2 = 2\,\mathrm{C}$. (2 pts)

3.
- (5 pts) For the pair of sheets shown in diagram (i), the displacement vectors at the origin due to charge sheet 1 and 2 can be written as

$$\begin{aligned}\mathbf{D}_1 &= \frac{1}{2}\rho_{s_1}\hat{x} = \frac{1}{2}\hat{x}\,\frac{\mathrm{C}}{\mathrm{m}^2}, \text{ and}\\ \mathbf{D}_2 &= \frac{1}{2}\rho_{s_2}(-\hat{x}) = -\frac{1}{2}\hat{x}\,\frac{\mathrm{C}}{\mathrm{m}^2},\end{aligned}$$

respectively. The resultant displacement vector will be $\mathbf{D} = \mathbf{D}_1 + \mathbf{D}_2 = 0$.

- (5 pts) Following the same procedure, the displacement vectors at the origin due to charge sheet 1 and 2 shown in diagram (ii) will be

$$\begin{aligned}\mathbf{D}_1 &= \frac{1}{2}\rho_{s_1}\hat{x} = \frac{1}{2}\hat{x}\,\frac{\mathrm{C}}{\mathrm{m}^2}, \text{ and}\\ \mathbf{D}_2 &= \frac{1}{2}\rho_{s_2}(-\hat{x}) = 1\hat{x}\,\frac{\mathrm{C}}{\mathrm{m}^2},\end{aligned}$$

respectively. The resultant displacement vector will be $\mathbf{D} = \mathbf{D}_1 + \mathbf{D}_2 = \frac{3}{2}\hat{x}\,\frac{\mathrm{C}}{\mathrm{m}^2}$.

4. This problem is similar to Example 4 in Lecture 3. The individual electric fields induced by the 3 slabs can be written as shifted-in-x versions of the result obtained in the example 2 of lecture 3. Knowing from Page 3 of Lecture 3 that the field of a constant volume density is given by $\mathbf{E} = \hat{x}\frac{\rho x}{\epsilon_o}$, we can write the electric field for each slab as (6 pts, 2 pts for each)

$$\mathbf{E}_1 = \begin{cases} -\hat{x}\frac{\rho_1}{2\epsilon_o}W = -\hat{x}\frac{\rho_o}{2\epsilon_o}W & \text{for } x < -2W \\ \hat{x}\frac{\rho_1}{\epsilon_o}\left(x + \frac{3}{2}W\right) = \hat{x}\frac{\rho_o}{2\epsilon_o}\left(2x + 3W\right) & \text{for } -2W < x < -W \\ \hat{x}\frac{\rho_1}{2\epsilon_o}W = \hat{x}\frac{\rho_o}{2\epsilon_o}W & \text{for } x > -W \end{cases}$$

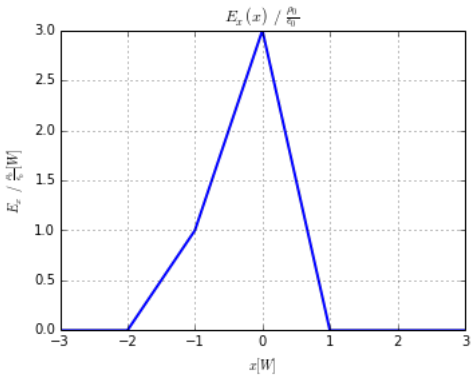
$$\mathbf{E}_2 = \begin{cases} -\hat{x}\frac{\rho_2}{2\epsilon_o}W = -\hat{x}\frac{\rho_o}{\epsilon_o}W & \text{for } x < -W \\ \hat{x}\frac{\rho_2}{\epsilon_o}\left(x + \frac{1}{2}W\right) = \hat{x}\frac{\rho_o}{\epsilon_o}\left(2x + W\right) & \text{for } -W < x < 0 \\ \hat{x}\frac{\rho_2}{2\epsilon_o}W = \hat{x}\frac{\rho_o}{\epsilon_o}W & \text{for } x > 0 \end{cases}$$

$$\mathbf{E}_3 = \begin{cases} -\hat{x}\frac{\rho_3}{2\epsilon_o}W = \hat{x}\frac{3\rho_o}{2\epsilon_o}W & \text{for } x < 0 \\ \hat{x}\frac{\rho_3}{\epsilon_o}\left(x - \frac{1}{2}W\right) = -\hat{x}\frac{3\rho_o}{2\epsilon_o}\left(2x - W\right) & \text{for } 0 < x < W \\ \hat{x}\frac{\rho_3}{2\epsilon_o}W = -\hat{x}\frac{3\rho_o}{2\epsilon_o}W & \text{for } x > W \end{cases}$$

Superposing the electric fields calculated for each slab,we obtain the total field (2 pts)

$$\mathbf{E} = \hat{x}E_x = \mathbf{E}_1 + \mathbf{E}_2 + \mathbf{E}_3 = \begin{cases} 0 & \text{for } x < -2W \\ \hat{x}\frac{\rho_0}{\epsilon_0}\left(x + 2W\right) & \text{for } -2W < x < -W \\ \hat{x}\frac{\rho_0}{\epsilon_0}\left(2x + 3W\right) & \text{for } -W < x < 0 \\ \hat{x}\frac{\rho_0}{\epsilon_0}\left(-3x + 3W\right) & \text{for } 0 < x < W \\ 0 & \text{for } x > W \end{cases},$$

which is plotted below. (2 pts)



5.

a) (4 pts) Curl and divergence of the vector field $\mathbf{F} = x\hat{x} + y\hat{y}$ are

$$\nabla \times \mathbf{F} = \nabla \times (x\hat{x} + y\hat{y}) = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & 0 \end{vmatrix} = 0,$$

and

$$\nabla \cdot \mathbf{F} = \left(\hat{x}\frac{\partial}{\partial x} + \hat{y}\frac{\partial}{\partial y} + \hat{z}\frac{\partial}{\partial z} \right) \cdot (x\hat{x} + y\hat{y}) = 2,$$

respectively. When we sketch the vector field $\mathbf{F}$, we obtain the following figure.

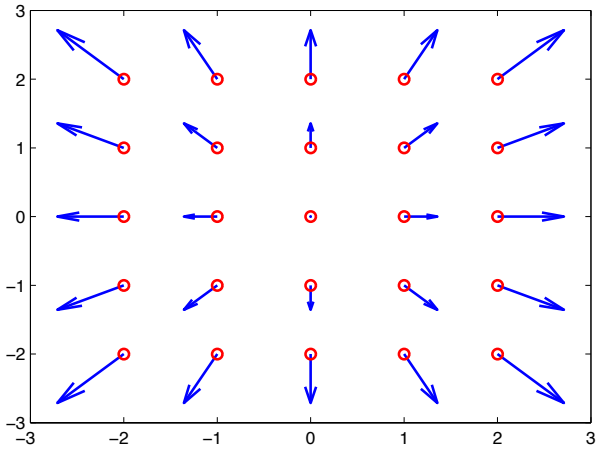


Figure 1: Vector field $\mathbf{F} = x\hat{x} + y\hat{y}$.

b) (4 pts) Applying the procedure in part (a) for $\mathbf{F} = y\hat{x} - x\hat{y}$, we obtain

$$\nabla \times \mathbf{F} = \nabla \times (x\hat{x} + y\hat{y}) = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y & -x & 0 \end{vmatrix} = -2\hat{z},$$

and

$$\nabla \cdot \mathbf{F} = \left(\hat{x}\frac{\partial}{\partial x} + \hat{y}\frac{\partial}{\partial y} + \hat{z}\frac{\partial}{\partial z} \right) \cdot (y\hat{x} - x\hat{y}) = 0$$

respectively. When we sketch the vector field $\mathbf{F}$, we obtain the following figure.

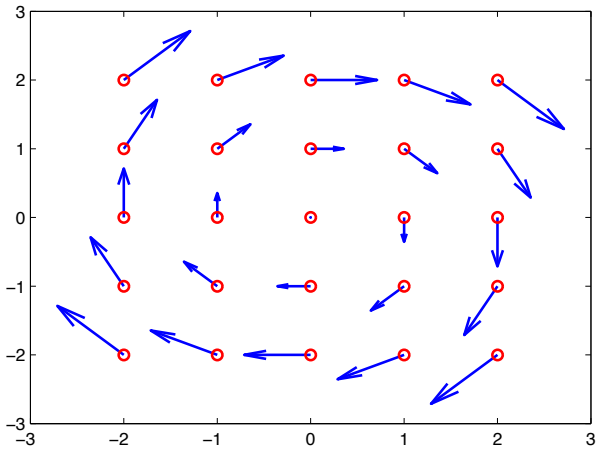


Figure 2: Vector field $\mathbf{F} = y\hat{x} - x\hat{y}$.

c) (2 pts)

i. **Curl** is a vector operator that describes the rotation of a 3-D vector field. In short, it is the vector representing circulation per unit area as seen in Figure 2. Since the vector field $\mathbf{F}$ in part (b) is defined only by the curl operator (i.e. divergence-free), $\nabla \times \mathbf{F} = -2\hat{z} \neq 0$ implies the field strength varies **across** the direction of the field.

ii. **Divergence** is an operator picking up the variation of the field strength along the direction of the vector field. As seen in part (b), the divergence operator picks up the field strength of the vector $\mathbf{F} = x\hat{x} + y\hat{y}$ which is curl-free. Therefore, $\nabla \cdot \mathbf{F} = \mathbf{2} \neq 0$ implies the field strength varies **along** the direction of the field.

6.

a) (5 pts) The static displacement field in the region is specified as $\mathbf{D} = -6\text{sgn}(z)\hat{z}\ \frac{\text{C}}{\text{m}^2}$. Using the differential form of Gauss' law, we determine

$$\rho = \nabla \cdot \mathbf{D} = \frac{\partial}{\partial z}(-6\text{sgn}(z)\,) = -12\delta(z)\,\frac{\text{C}}{\text{m}^3},$$

where $\delta(\cdot)$ is the Dirac delta function and $\frac{\partial}{\partial z}(\text{sgn}(z)) = 2\delta(z)$. Then, the displacement field generated by an infinite surface charge ρ_s along the xy-plane can be written as

$$\mathbf{D} = \begin{cases} \frac{\rho_s}{2}\,\hat{z} & z > 0 \\ -\frac{\rho_s}{2}\,\hat{z} & z < 0 \end{cases} = \frac{\rho_s}{2}\text{sgn}(z)\,\hat{z}.$$

Referring to pages 6 and 7 in Lecture 3, we can state that an infinite surface charge density $\rho_s = -12\ \frac{\text{C}}{\text{m}^2}$ can be represented as a volumetric charge density $\rho = \rho_s\delta(z) = -12\delta(z)\ \frac{\text{C}}{\text{m}^3}$.

b) (5 pts) Given the static displacement field given in the problem, we find

$$\rho = \nabla \cdot \mathbf{D} = \frac{\partial D_z}{\partial z} = \begin{cases} 0 & z < -4\,\mathrm{m}, \\ 4\,\frac{\mathrm{C}}{\mathrm{m}^2} & |z| < 4\,\mathrm{m}, \\ 0 & z > 4\,\mathrm{m}. \end{cases}$$

7. Taking the curl of the Coulomb's field of a charge Q, we find (2 pts)

$$\nabla \times \mathbf{E} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & E_z \end{vmatrix} = \left(\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z}\right)\hat{x} + \left(\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x}\right)\hat{y} + \left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y}\right)\hat{z},$$

whose each component is solved as (6 pts)

$$\begin{aligned} \frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} &= \frac{Q}{4\pi\epsilon_o}\left[\frac{\partial}{\partial y}\left(\frac{z}{r^3}\right) - \frac{\partial}{\partial z}\left(\frac{y}{r^3}\right)\right] \\ &= -\frac{Q}{4\pi\epsilon_o}\left(\frac{3z}{r^4}\frac{\partial r}{\partial y} - \frac{3y}{r^4}\frac{\partial r}{\partial z}\right) \\ &= -\frac{Q}{4\pi\epsilon_o}\left(\frac{3zy}{r^5} - \frac{3yz}{r^5}\right) = 0, \end{aligned}$$

likewise,

$$\begin{aligned} \frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x} &= \frac{Q}{4\pi\epsilon_o}\frac{\partial}{\partial z}\left(\frac{x}{r^3}\right) - \frac{Q}{4\pi\epsilon_o}\frac{\partial}{\partial x}\left(\frac{z}{r^3}\right) = 0, \\ \frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} &= \frac{Q}{4\pi\epsilon_o}\frac{\partial}{\partial x}\left(\frac{y}{r^3}\right) - \frac{Q}{4\pi\epsilon_o}\frac{\partial}{\partial y}\left(\frac{x}{r^3}\right) = 0. \end{aligned}$$

Therefore, the curl of the electrostatic field generated by a point charge is always zero, i.e. $\nabla \times \mathbf{E} = 0$. (2 pts)